Graduation Project: Geo-Bound Crypto-Vault & DevSecOps Pipeline

Discussion Work Document - Student 3 Tasks Plan (Geo-Fencing & Spatial Verification)

Academic Supervisor:	Dr. Rami Matarneh	**Security Domain:**	Geo-Bound Access & Access Control
Student 3 Role:	Geo-Fencing Mechanism & Distance Calculation	**Tech Stack:**	Python, Haversine Formula, GeoIP / GPS

1. Scope of Work & Core Responsibility

Student 3 is responsible for designing and building the **Geo-Bound Access Layer** for the encrypted vault system. This layer aims to enforce spatial security controls by preventing data decryption operations unless the requesting user is physically located within a designated geographic perimeter, serving as an essential security layer alongside encryption and user passwords.

2. Core Technical Architecture Elements

- **Geo-Fence Zone Definition:** Defining target central coordinates (Target Latitude & Longitude) and a allowed radius threshold (e.g., 50m / 100m).
- **Spatial Distance Calculation:** Programming a verification function based on the *Haversine Formula* to accurately compute the spherical distance between the user's real-time location and the target perimeter.
- **Access Control Rules:** Enforcing dynamic access control decisions automatically based on the computed distance metric.

3. Mathematical Formulation & Algorithmic Logic (Haversine)

The Haversine formula calculates the great-circle distance between two points on a sphere given their longitudes and latitudes:

```
a = sin²(Δφ/2) + cos(φ1) · cos(φ2) · sin²(Δλ/2) c = 2 · atan2( √a, √(1−a) ) d = R · c (where R = 6,371,000 meters)
```

4. Access Control Decision Matrix

Scenario / Condition	Programmatic Logic	Action Taken
User Inside Perimeter	Distance ≤ Allowed Radius	**Grant Access:** Pass key to decryption module
User Outside Perimeter	Distance > Allowed Radius	**Deny Access:** Return "Out of Geo-Fence Boundaries" error
GPS Permission Denied	Location Data == Null	**Deny Access:** Enforce Zero-Trust default policy

Scenario / Condition	Programmatic Logic	Action Taken
Spoofing Detected	Inconsistent Mock Location	**Deny & Log Alert:** Record suspicious attempt

5. Risk Mitigation & Addressing Supervisor Feedback

> **Clarification addressing Supervisor Note #4:**
> Relying solely on GPS can be susceptible to location manipulation (GPS Spoofing). Therefore, the following mitigation strategies fall directly under Student 3's implementation scope:

- **Hybrid Location Verification:** Cross-referencing GPS coordinates with network-based IP Geolocation to detect discrepancies.
- **Mock Location Detection:** Immediately rejecting requests if mock location providers or spoofing applications are detected on the client device.
- **Fail-Closed Policy:** If location permissions are revoked or GPS signal accuracy is degraded, the system defaults to an immediate denial of access in line with Zero-Trust principles.